



# TM1 Review Guide for Test 1 on 02.12

## Completion requirements

**Test format.** Test 1 will be administered at the start of class. You will have 30 minutes to complete the test. It will consist of short answer and simple problem solving. You may occasionally encounter multiple choice questions. After the test, we will continue working on the first lab.

- General topics:
  - **What is a data scientist?** Answer that question as you would for another student.
  - **What is *Big Data*?** In what ways does it differ from other types of data?
  - One of our textbooks (PDSH) includes a Venn diagram describing *data science* as an interdisciplinary discipline. That diagram includes an area called the "**danger zone**." What do you think that means? Why is it dangerous? Then go and look up [the original link in a blog article by Drew Conway](#). **Read why Conway describes it as a danger zone.** Do you agree or disagree with his assessment? Explain.
  - **Describe several roles of people in an organization *other than the data scientist*** who may be working on a data analytics team or otherwise vital to the success of an analytics project. Briefly discuss how you would relate to persons in each role for the success of a project.
  - **In which phase would the team expect to invest most of the project time?** Why? Where would the team expect to spend the **least time**? Explain.
  - Data analytics frameworks such as Jupyter Notebook and R Studio are sometimes described as "**analytic sandboxes**." However, the description of "analytic sandbox" in our EMC textbook [Chs. 1-2] seems to go beyond this. According to the discussion in the book, what is an analytic sandbox, and why is it important?
  - Imagine that you have just been hired as a data scientist for the Acme Co., a startup. Briefly, describe some of your tasks on a typical day. Also describe some of the challenges that you face.
  - **What is BI? Explain the differences between BI and Data Science.**
- Give an overview of the **data analytics lifecycle** as described in our EMC textbook (ch. 2). Include a 1-2 sentence of each phase. Note: I won't often ask you to memorize a list. But this is one you should know by memory.
- **Data science team roles.** Briefly compare and contrast the following roles.
  - **Data Scientist:** A data scientist is a person who works with data to extract insights and make predictions. They use a variety of methods and

technologies including machine learning, data mining, and statistical analysis. They need to possess strong analytical, problem-solving, communication, and programming skills.

- **Machine Learning Engineer:** A machine learning engineer is a person who designs and builds machine learning models. They use a variety of technologies, including deep learning frameworks and natural language processing. They need to possess strong analytical, problem-solving, communication, and programming skills.
- **Data Engineer:** A data engineer is a person who designs, builds, and maintains data systems. They need to possess strong problem solving and programming skills.
- **DB Administrator:** A database administrator is a person who is responsible for managing and maintaining databases. They need to possess strong problem solving and programming skills.
- **Software Engineer:** A software engineer is a person who designs and builds software systems. They need to possess strong problem solving and programming skills.
- **Statistician:** A statistician is a person who applies statistical methods to data. They use a variety of techniques, including probability theory and data analysis. They need to possess strong analytical and communication skills.
- **BI Analyst:** A business intelligence analyst is a person who works with data to analyze and report on business performance. They use a variety of methods, including data visualization, to uncover insights and identify trends. They need to possess strong analytical, problem-solving, communication, and data visualization skills.
- **Project manager:** A project manager is a person who is responsible for managing the project, including budgeting, scheduling, and resource allocation. They need to possess strong communication and organizational skills.
- **Project sponsor:** A project sponsor is a person who provides the financial and moral support for the project. They need to possess strong communication and organizational skills.
- Make sure you know what each of these tools does:
  - **Python3:** is a programming language used for creating scripts and applications. It is popular for data science and machine learning, and is often used in conjunction with other libraries.
  - **Jupyter Notebook:** is an interactive coding environment used for writing and running code, as well as presenting data visualizations.
  - **Pandas:** is a data manipulation and analysis library used for working with tabular data, such as data frames and series.

- **NumPy:** is a library for scientific computing that provides tools for working with arrays and matrices, and is often used for linear algebra.
- **Matplotlib:** is a library for creating static, animated, and interactive visualizations. It can be used in conjunction with other libraries like Pandas and Scikit-learn.
- **Seaborn:** is a library for creating attractive statistical data visualizations. It is built on top of Matplotlib and is often used in conjunction with other libraries.
- **Scikit-learn (sklearn):** is a library for machine learning that provides tools for training, testing, and evaluating models. It can be used with other libraries like NumPy and Pandas.
- **Markdown:** is a lightweight markup language used for formatting plain text documents, such as README files. It is often used in conjunction with other tools, like Jupyter Notebook.
- **LaTeX:** is a typesetting system used for creating documents, such as scientific papers, that requires a high degree of precision and accuracy. It is used to create high-quality documents with mathematical symbols and equations. It can be used within Jupyter Notebook.
- Jupyter Notebook questions:
  - What is the difference between a **code cell** and a **Markdown cell**? How do you change a cell from one type to the other?
  - What is the purpose of a Jupyter Notebook **kernel**?

*Note: There will be no code-specific problems on the test this year. There will be a coding quiz that will help evaluate skills you should have developed in CSC 170 and DSC 270. This coding quiz will NOT count toward your course grade. It is only to help clarify your coding skills entering the course.*

Last modified: Monday, February 24, 2025, 1:40 PM